

Jordan Newman

Software Engineer | Desktop Products, Systems, Agentic AI

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Willing and able to work in person in New York City

SUMMARY

Product-minded senior software engineer with more than 20 years of experience delivering commercial application, integration, systems, and infrastructure features. Currently building Nexus, a local-first agentic control plane with a React Web UI, Tauri desktop client, multi-agent workflows, default-deny tool permissions, human approvals, layered knowledge, and auditable execution. Works holistically across the stack and uses profiling, iterative measurement, and architecture tradeoffs to solve technical problems. Hands-on with React, TypeScript, Electron, Tauri, C#, C++, and C, plus two-plus years of ML-framework and local-model experimentation with LM Studio, llama.cpp, Ollama, and LM Studio Bionic.

TECHNICAL SKILLS

- **Application Engineering:** React, TypeScript, Tailwind CSS, Tauri, Electron, C#, C++, C, Rust, Python, Objective-C, desktop applications, cross-stack development
- **Agentic AI and Local Models:** agent workflows, tool calling, human-in-the-loop approvals, model routing, MCP, LM Studio, llama.cpp, Ollama, LM Studio Bionic, MLX, Hugging Face Transformers, TensorFlow
- **Systems and Data:** Linux, FreeBSD, SQLite, local-first architecture, process supervision, server-sent events, HTTP, TCP/IP, non-blocking I/O, MultiValue/UniVerse
- **Security and Delivery:** default-deny permissions, OAuth 2.0, operating-system keyrings, audit logging, GitHub, CI/CD, Docker, Kubernetes, Terraform, Azure DevOps
- **Debugging and Performance:** GDB, Valgrind, Linux perf, CPU profiling and optimization, system design, architecture

SELECTED PRODUCT AND SYSTEMS ENGINEERING

Nexus Agentic Operating System — Creator and Engineer

Active personal project

- Designed and built a local-first agentic control plane that persists tasks, approvals, audit records, and knowledge in SQLite and the local filesystem before introducing cloud dependencies.
- Implemented declarative YAML, DAG-oriented workflows with sub-tasks, parallel agents, conditional branches, controlled loop-back execution, and audited task-state transitions.
- Enforced default-deny tool permissions through risk levels, mandatory and standing approvals, workspace grants, air-gapped network controls, canonical paths, and symlink-traversal protection.
- Crafted React/Tailwind product experiences for task creation, live events, approvals, artifacts, workflow visualization, knowledge exploration, AI-usage statistics, routing simulation, and configuration.
- Built a Tauri desktop client with system-tray controls, workspace selection, notifications, keychain-backed secrets, file and URL opening, and local process supervision.
- Migrated from Electron to Tauri for a smaller footprint, faster startup, lower memory use, and a stricter capability-based security model.
- Implemented model routing with health checks, circuit breakers, budget-aware selection, failover, prompt caching, and privacy-conscious token, latency, retry, error, and cost tracking.

Open-Source Engineering

- Contributed a merged Rust Azure Key Vault provider to cachix/secretspec, covering multiple authentication modes, validation, integration tests, sovereign-cloud support, documentation, and maintainer-review fixes.

PrimeHTTPD and CDN Platform

- Architected PrimeHTTPD, a non-blocking HTTP/CDN server in C for FreeBSD using kqueue, blocking-I/O workers, Unix descriptor passing, shared-memory scheduling, and zero-copy sendfile().
- Deployed PrimeHTTPD across approximately 200 servers and more than 150,000 concurrent connections within infrastructure spanning 3,000-4,000 servers, 10 locations, and over 65 Gbps.

PROFESSIONAL EXPERIENCE

Advantive — Senior Development Specialist, APIs and Integrations

Remote / Tampa Bay, FL | June 2022 - Present

- Contribute to Architecture Team decisions, engineering standards, platform direction, reusable libraries, documentation, developer tooling, and AI-assisted engineering workflows.
- Implement and improve Azure DevOps, Docker, Kubernetes, Terraform, CI/CD, and security-scanning workflows across engineering teams.

DDI System — Senior Development Specialist

Manalapan, NJ | October 2021 - July 2022

- Developed and maintained Inform's primarily C# desktop applications, including WebCom and PrintCom connectivity to a UniVerse database backend.
- Added OAuth 2.0 support for email and implemented SellerCloud and product-catalog integrations.

ISPRIME — Chief Executive Officer

Weehawken, NJ | March 2018 - December 2019

- Led datacenter modernization, operational restructuring, and improvements to internally developed monitoring, security, and infrastructure platforms.

DDI System — Computer Programmer

Manalapan, NJ | April 2017 - March 2018

- Built ERP integrations and automated order, catalog, image, safety-data, SFTP, SellerCloud, supplier, and MultiValue/UniVerse workflows.

Too Much Media LLC — Computer Programmer

Morganville, NJ | January 2016 - April 2017

- Developed real-time browser chat integrated with ticketing workflows and implemented OAuth 2.0 authorization for protected customer sites.

MFCXY, Inc. — Computer Programmer

Chicago, IL | December 2014 - September 2015

- Developed backend and Windows desktop-client features plus middleware for cross-platform database migration.

ISPRIME — Owner and CIO

Weehawken, NJ | January 2001 - December 2014

- Architected and operated the production hosting and CDN platform and personally built its HTTP serving, kernel, networking, security, monitoring, authentication, and operational software.
- Created training documentation, trained the majority of employees over the years, and mentored two beginners over several years into technical experts and company leaders.

ADDITIONAL EXPERIENCE

AJPM, LLC — Programmer / Owner | November 2012 - December 2013

FatWallet.com — Systems Administrator | December 2000 - June 2001

EDUCATION

Brookdale Community College — Computer Science coursework; left during the first semester to pursue ISPRIME full-time during rapid company growth.